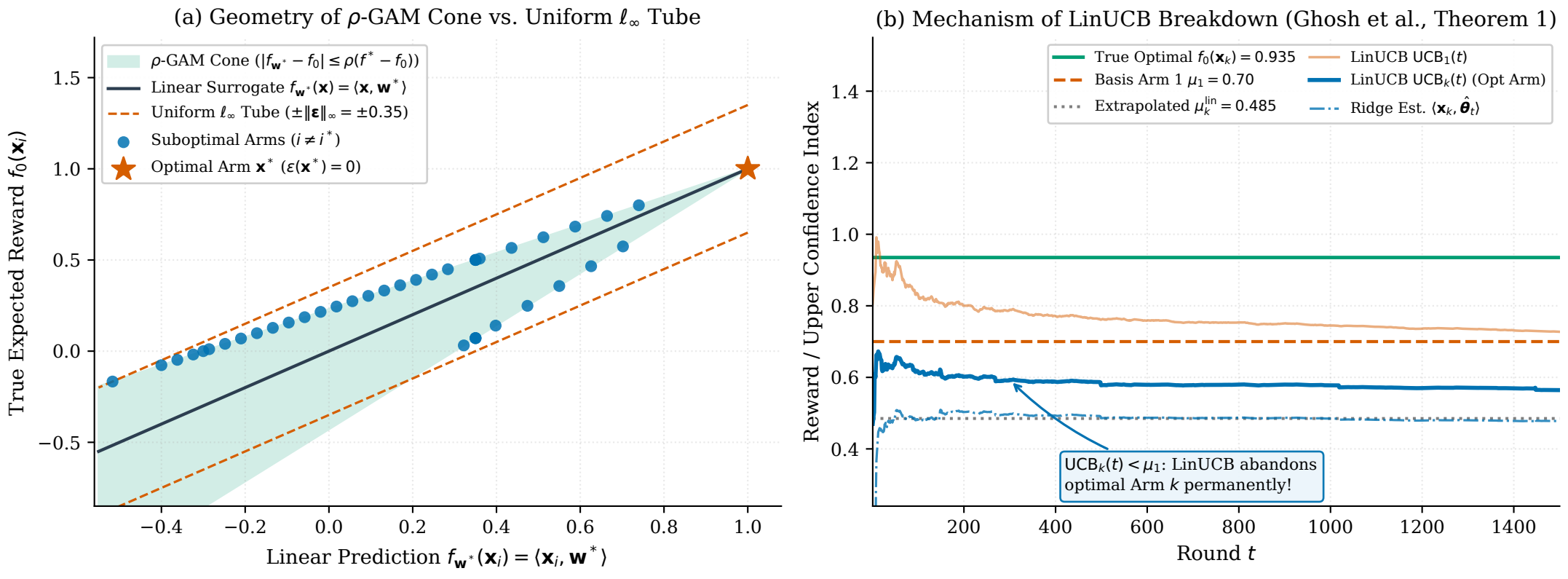

(a) Geometry of ρ-GAM Cone vs. Uniform ℓ_∞ Tube
ρ-GAM Cone ($|f_{\mathbf{w}^*} - f_0| \le \rho(f^* - f_0)$)
Linear Surrogate $f_{\mathbf{w}^*}(\mathbf{x}) = \langle \mathbf{x}, \mathbf{w}^* \rangle$
Uniform ℓ_∞ Tube ($\pm\|\boldsymbol{\varepsilon}\|_\infty = \pm 0.35$)
Suboptimal Arms ($i \neq i^*$)
Optimal Arm $\mathbf{x}^*$ ($\varepsilon(\mathbf{x}^*) = 0$)
True Expected Reward $f_0(\mathbf{x}_i)$
Linear Prediction $f_{\mathbf{w}^*}(\mathbf{x}_i) = \langle \mathbf{x}_i, \mathbf{w}^* \rangle$
1.5
1.0
0.5
0.0
−0.5
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
(b) Mechanism of LinUCB Breakdown (Ghosh et al., Theorem 1)
True Optimal $f_0(\mathbf{x}_k) = 0.935$
Basis Arm 1 $\mu_1 = 0.70$
Extrapolated $\mu_k^{\text{lin}} = 0.485$
LinUCB $\text{UCB}_1(t)$
LinUCB $\text{UCB}_k(t)$ (Opt Arm)
Ridge Est. $\langle \mathbf{x}_k, \hat{\boldsymbol{\theta}}_t \rangle$
Reward / Upper Confidence Index
$\text{UCB}_k(t) < \mu_1$: LinUCB abandons optimal Arm k permanently!
Round t
1.4
1.2
1.0
0.8
0.6
0.4
200
400
600
800
1000
1200
1400